

netan QGIS Plugin

This plugin wraps the local netan CLI from inside QGIS.

It is meant for the current repository layout:

- one local workspace root
- one `.netan/` state directory
- one netan executable built from this repo

The plugin is now written to be usable in both QGIS 3 and QGIS 4, and it supports two execution modes:

- Native: QGIS and netan run on the same OS
- WSL: Windows QGIS launches the Linux netan binary through `wsl.exe`

What It Does

- refreshes dataset manifests from `.netan/datasets/`
- validates and compiles profiles against the selected dataset
- writes route request JSON from typed coordinates
- runs route, od, and matrix analyses through the CLI
- loads `.geojson`, `.gpkg`, and `.geoparquet` outputs into QGIS after a successful run

Install In QGIS

1. Build the CLI:

```
cargo build -p netan-cli
```

1. Copy or symlink `netan_qgis` into your QGIS profile plugin directory.

Linux example:

```
mkdir -p ~/.local/share/QGIS/QGIS3/profiles/default/python/plugins
ln -s /home/elmeriniemi/stuff/netan/qgis_plugin/netan_qgis
~/.local/share/QGIS/QGIS3/profiles/default/python/plugins/
netan_qgis
```

Windows QGIS 4 example:

```

New-Item -ItemType Directory -Force
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins"
cmd /c mklink /D
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins\netan_qgis"
"\\wsl$
\Ubuntu\home\elmeriniemi\stuff\netan\qgis_plugin\netan_qgis"

```

1. In QGIS, enable the plugin from Plugins -> Manage and Install Plugins.

First Run

Set these fields in the dock:

- Execution mode: Native or WSL
- QGIS-visible workspace root:
 - Linux native example: /home/elmeriniemi/stuff/netan
 - Windows + WSL example: \\wsl\$
 - \Ubuntu\home\elmeriniemi\stuff\netan
- Native netan executable for native mode, usually /home/elmeriniemi/stuff/netan/target/debug/netan
- WSL distro, WSL workspace root, and WSL netan executable for WSL mode
- Dataset: choose an imported dataset after pressing Refresh
- Profile: point at a YAML/TOML profile such as examples/profiles/car_research_v1.yml

If the dataset has not been imported yet, do that once from the CLI or the native GUI first.

Windows QGIS + WSL

Yes, you can use this from Windows QGIS even if you build and run netan inside WSL.

Use these settings:

- Execution mode: WSL
- QGIS-visible workspace root: \\wsl\$
 - \Ubuntu\home\elmeriniemi\stuff\netan
- WSL distro: Ubuntu
- WSL workspace root: /home/elmeriniemi/stuff/netan
- WSL netan executable: /home/elmeriniemi/stuff/netan/target/debug/netan

How it works:

- QGIS reads manifests and output layers through the Windows-visible \\wsl\$ path.

- The plugin launches `wsl.exe` and runs the Linux `netan` binary inside your distro.
- Relative paths like `examples/profiles/car_research_v1.yml` or `.netan/runs/qgis-route.geojson` are translated automatically.

This is the simplest path if your repo and build artifacts live inside WSL.

If you want the cleanest Windows integration, build a native Windows `netan.exe` and use Native mode instead.

Full Windows Example

This example shows both supported Windows flows for QGIS 4.

A. Windows QGIS 4 calling into WSL

1. Build the CLI in WSL:

```
cd /home/elmeriniemi/stuff/netan
cargo build -p netan-cli
```

1. Optionally copy the repo to Windows without the Rust build directory:

```
cd /home/elmeriniemi/stuff/netan
mkdir -p /downloads/netan
for p in .* *; do
    [ "$p" = "." ] || [ "$p" = ".." ] || [ "$p" = "target" ] || cp -
a -- "$p" /downloads/netan/
done
```

1. Install the plugin into the Windows QGIS 4 profile:

```
New-Item -ItemType Directory -Force
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins"
cmd /c mklink /D
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins\netan_qgis"
"\\wsl$
\Ubuntu\home\elmeriniemi\stuff\netan\qgis_plugin\netan_qgis"
```

1. Open QGIS 4 and enable the plugin.

2. Configure the dock with:

- Execution mode: WSL
- QGIS-visible workspace root: `\\wsl$
\Ubuntu\home\elmeriniemi\stuff\netan`
- WSL distro: Ubuntu
- WSL workspace root: `/home/elmeriniemi/stuff/netan`

- WSL netan executable: /home/elmeriniemi/stuff/netan/target/debug/netan
- Profile: examples/profiles/car_research_v1.yml

1. Press Refresh, select groningen_2026_03, then run:

- route output: .netan/runs/qgis-route.geojson
- OD output: .netan/runs/qgis-od.gpkg
- matrix output: .netan/runs/qgis-matrix.gpkg

B. Windows QGIS 4 with native netan.exe

1. Copy the repo to Windows:

```
cd /home/elmeriniemi/stuff/netan
mkdir -p /windowdownloads/netan
for p in .* *; do
    [ "$p" = "." ] || [ "$p" = ".." ] || [ "$p" = "target" ] || cp -
a -- "$p" /windowdownloads/netan/
done
```

1. Build the Windows executable from PowerShell:

```
cd C:\path\to\netan
cargo build -p netan-cli
```

1. Install the plugin:

```
New-Item -ItemType Directory -Force
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins"
cmd /c mklink /D
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins\netan_qgis"
"C:\path\to\netan\qgis_plugin\netan_qgis"
```

1. Configure the dock with:

- Execution mode: Native
- QGIS-visible workspace root: C:\path\to\netan
- Native netan executable: C:\path\to\netan\target\debug\netan.exe
- Profile: examples/profiles/car_research_v1.yml

1. Import data and compile a profile if needed:

```
cd C:\path\to\netan
.\target\debug\netan.exe dataset import
datasets\groningen-260317.osm.pbf --name groningen_2026_03
.\target\debug\netan.exe profile compile --dataset
groningen_2026_03 --profile examples\profiles\car_research_v1.yml
```

1. Use the plugin normally from QGIS 4.

Example Flows

Route

1. Open the plugin dock.
2. Set the execution mode and the matching workspace/executable fields.
3. Press Refresh.
4. Pick groningen_2026_03 as the dataset.
5. Set profile to examples/profiles/car_research_v1.yml.
6. Press Compile Profile.
7. In the Route tab, set:
 - request path: examples/requests/route_from_qgis.json
 - output path: .netan/runs/qgis-route.geojson
 - origin: 6.5665, 53.2194
 - destination: 6.5716, 53.2148
8. Press Write Request.
9. Press Run Route.

The resulting line layer loads into QGIS automatically.

OD

Use:

- pairs path: examples/requests/od_pairs.csv
- output path: .netan/runs/qgis-od.gpkg

Then press Run OD.

Matrix

Use:

- origins path: examples/requests/matrix_origins.csv
- destinations path: examples/requests/matrix_destinations.csv
- output path: .netan/runs/qgis-matrix.gpkg

Then press Run Matrix.

Notes

- The plugin does not embed the Rust engine; it shells out to the local CLI.
- Relative paths are resolved against the configured workspace root for the active mode.
- Only spatial outputs are auto-loaded into QGIS.
- In WSL mode, output layers are loaded back into QGIS through `\\wsl$\\<distro>\\....`